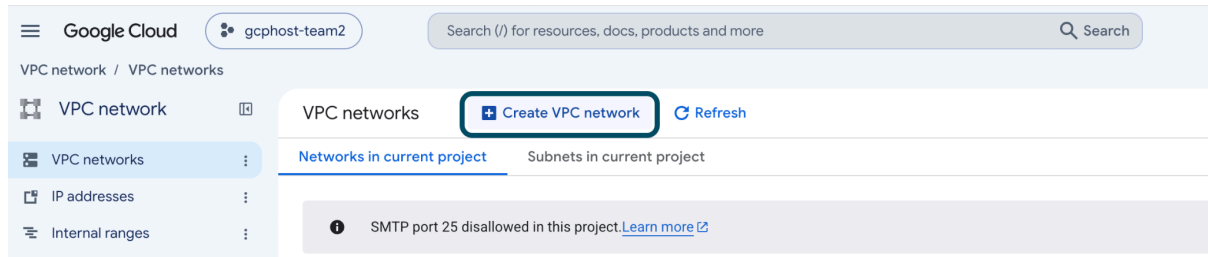


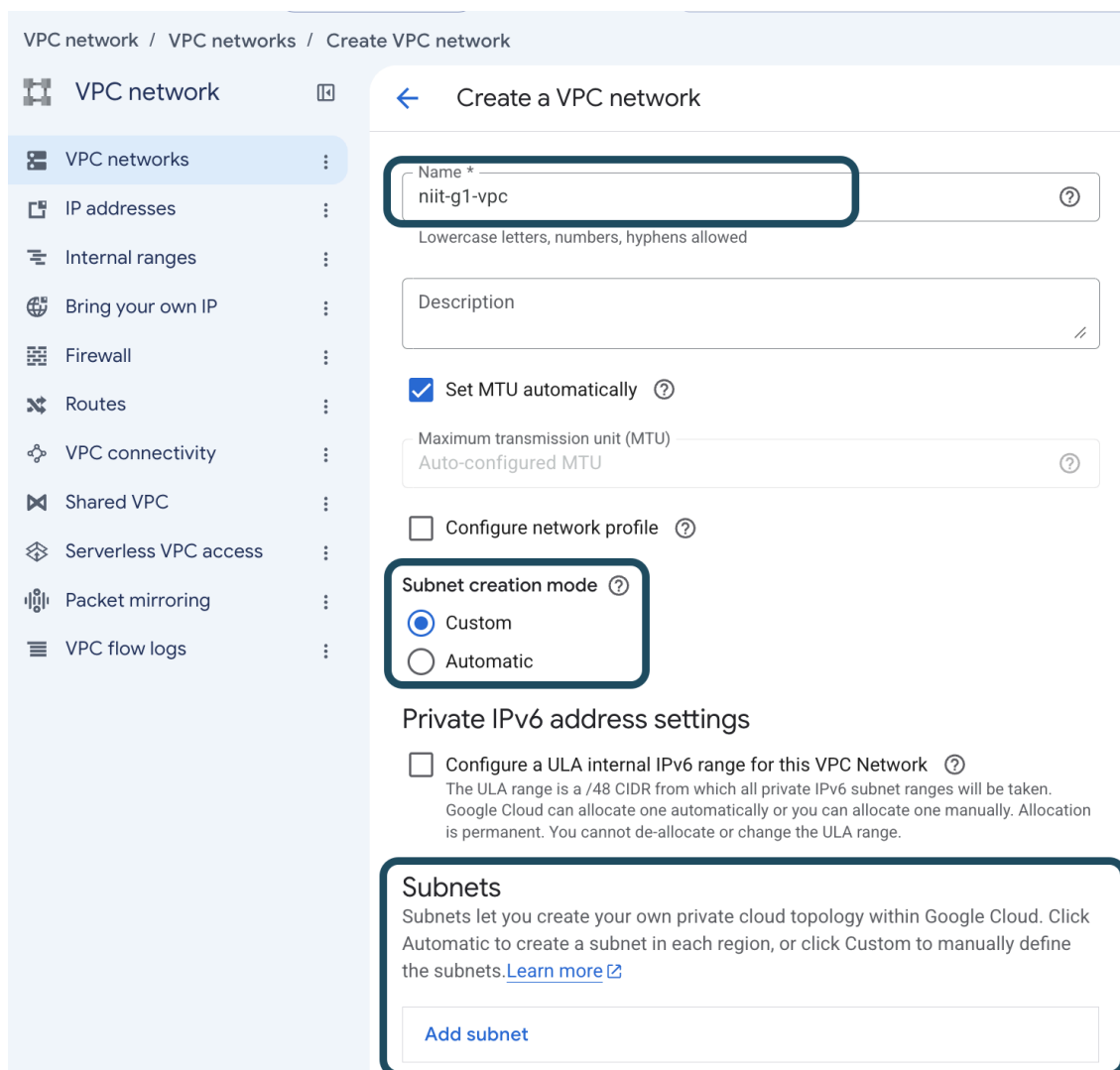
GCP VPC Setup

Step 1: VPC Creation *[to be performed by only mentor or team lead]*

1. Go to VPC Network > Create New VPC Network



2. Input name as niit-t1-vpc. Use placeholder t1, t2, t3 as per the project allocation



3. Keep everything else as default.
4. As subnet creation mode is custom, it might have an input field for subnet. Delete the same.
5. Create VPC

Step 2: Subnet Creation - [to be performed by all pax individually]

1. Go to VPC Network > locate niit-<t1>-vpc. (t1 might be replaced with your project relevant naming)

VPC network / VPC networks

VPC network + Create VPC network ↻ Refresh

Networks in current project Subnets in current project

niit-g1-vpc

Name	Subnets	MTU	Mode	IPv6 ULA range	Gateways	Flow log configs	Firewall rules	Global dynar
default	43	1460	Auto				6	Off
gcphost-team2-postgres-lab-vpc	1	1460	Custom				2	Off
niit-g1-vpc	0	1460	Custom				0	Off

2. Notice we have 0 subnets here.
3. Let's create a subnet for deployment purposes. Click on VPC name.
4. Go to **Subnet** tab and click on **Add Subnet**. It shall open a sidepane towards right.

VPC network / VPC networks / Network: niit-g1-vpc

VPC network details ✕ Delete VPC network 🕒 History ℹ️ Show info panel

niit-g1-vpc

< Overview **Subnets** Static internal IP addresses Firewalls Endpoint groups Firewall endpoints Routes VPC network peering Priv >

Subnets + Add subnet ☰ Manage flow logs

Filter Enter property name or value

Name	Region	Stack type	Primary IPv4 range	Secondary IPv4 ranges	IPv6 ranges	Reserved internal ranges	PDP
No rows to display							

Reserved proxy-only subnets for load balancing

Name	Region	IP address ranges	Gateway	Role	Purpose
No rows to display					

[Equivalent REST](#)

5. Input details as follows:
 - a. Name: your_email_username eg niit1, niit4
 - b. Region: eg. us-east5.
 - i. Note: All in a project pax are advised to use different regions. Refer to the table in instruction 6 for recommended regions

- c. IPv4 range: 10.202.0.0/20
 - i. Note: Use range that is recommended with the relevant region in the table under step 6
- d. Set **Private Google Access** to **On**
- e. Create Subnet

The screenshot shows the 'Add a subnet' dialog in the Google Cloud Platform console. The dialog is for the VPC network 'niit-g1-vpc'. The 'Name' field is set to 'niit4'. The 'Region' is set to 'us-east5 (Columbus)'. The 'Purpose' is set to 'None'. The 'IP stack type' is set to 'IPv4 (single-stack)'. The 'Primary IPv4 range' is set to '10.202.0.0/20'. The 'Associate with an internal range' checkbox is unchecked.

6. Subnet region & CIDR recommendation table

Region	Ipv4 CIDR range
us-central1	10.128.0.0/20
us-west1	10.138.0.0/20
us-east5	10.202.0.0/20
us-east1	10.142.0.0/20

7. Once the subnet is create, you can verify the same under **Subnet** tab in your VPC.

The screenshot shows the 'Subnets' tab in the Google Cloud Platform console for the VPC network 'niit-g1-vpc'. The table lists the subnet 'niit4' with the following details:

Name	Region	Stack type	Primary IPv4 range	Secondary IPv4 ranges	IPv6 ranges	Reserved internal ranges
niit4	us-east5	IPv4 (single-stack)	10.202.0.0/20			None

Step 3: Create firewall rules

1. Click on Firewall tab under your VPC

VPC network details [Delete VPC network](#) [History](#)

niit-g1-vpc

< Overview Subnets Static internal IP addresses **Firewalls** Endpoint groups Firewall endpoints Routes VPC network peering P >

Policy view Firewall rule view Mirroring rule view

Hierarchical policies [Refresh](#)

Filter Enter property name or value

Some of the data is not available due to lack of permissions

Name	Enforcement order	Deployment scope	Firewall rules
No rows to display			

VPC firewall rules [+ Create VPC firewall rule](#) [Refresh](#)

Filter Enter property name or value

Name	Enforcement order	Deployment scope	Priority	Description	Direction of traffic	Target	Source	Destination	Protocols
No rows to display									

2. Set configurations as follows:

- Name: niit4-cluster-internal (use your username)
- Network: niit-<t1>-vpc (use your project specific VPC)
- Priority: 1000
- Direction of traffic: Ingress
- Action on match: Allow
- Specified target tags: niit4-node
- Ipv4 ranges: 10.202.0.0/20 (use range of your subnet)
- Protocols
 - tcp: 2379, 2380, 5432, 8008
 - other : icmp
- Create firewall rule

← Create a firewall rule

Firewall rules control incoming and outgoing traffic to an instance. By default, all incoming traffic to your network is blocked. [Learn more](#)

Name *

niit4-cluster-internal



Lowercase letters, numbers, hyphens allowed

Description



Logs

Turning on firewall logs can generate a large number of logs, which can increase costs in Logging. [Learn more](#)



On



Off

Network *

niit-g1-vpc



Priority *

1000

[Compare](#)



Priority can be 0–65535

Direction of traffic ?



Ingress



Egress

Action on match ?



Allow



Deny

Targets

Specified target tags



Target tags *

niit4-node



← Create a firewall rule

Source filter

IPv4 ranges



Source IPv4 ranges *

10.202.0.0/20



for example, 0.0.0.0/0, 192.168.2.0/24



Second source filter

None



Destination filter

None



Protocols and ports ?



Allow all



Specified protocols and ports

☒ TCP

Ports

2379,2380,5432,8008

E.g. 20, 50–60

☐ UDP

Ports

E.g. all

☐ SCTP

Ports

E.g. 20, 50–60

☒ Other

Protocols *

icmp

Separate multiple protocols by commas, e.g. AH, ICMP

Similarly, create another firewall rule for SSH with following configurations:

- j. Name: niit4-iap-ssh (use your username)
- k. Network: niit-<t1>-vpc (use your project specific VPC)
- l. Priority: 1000
- m. Direction of traffic: Ingress
- n. Action on match: Allow
- o. Specified target tags: niit4-ssh
- p. Ipv4 ranges: your public access ip (use range of your subnet)
- q. Protocols
 - i. tcp: 22
- r. Create firewall rule